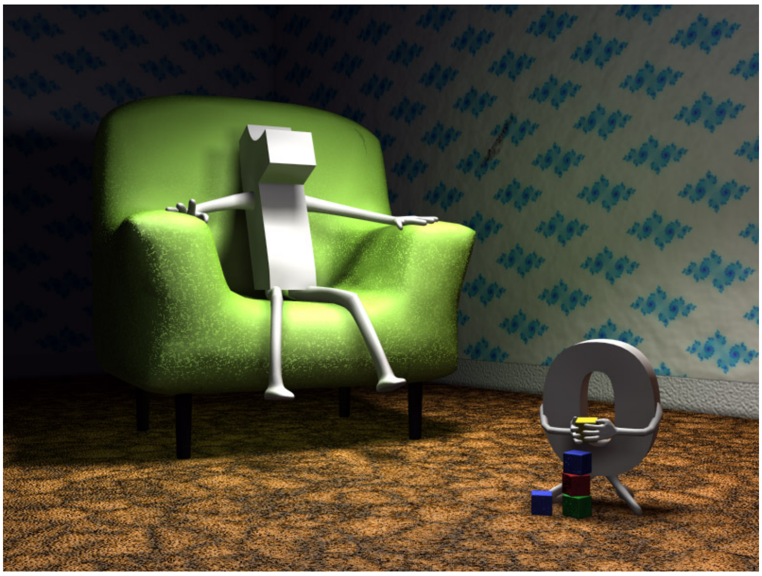
2IA Practice Problems I

Integers



Practice Problems Chapter 1

1. **Prove each statement by induction on n.**

(a) $1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$, for all $n \geq 1$.

Solution: Given an integer $n \geq 1$, let us consider the following statement

$$p_n : \quad 1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Notice that p_1 is true, since

$$\frac{1(1+1)(2 \cdot 1 + 1)}{6} = 1$$

Take $k > 1$, and assume that p_k is true. We then need to show that p_{k+1} is true, namely

$$\begin{aligned} 1^2 + 2^2 + \ldots + k^2 + (k+1)^2 &\stackrel{?}{=} \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6} = \\ &= \frac{(k+1)(k+2)(2k+3)}{6} \end{aligned}$$

Apply the induction hypothesis to get that

$$\begin{aligned} &1^2 + 2^2 + \ldots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \\ &= \frac{1}{6}(k+1)\big(k(2k+1) + 6(k+1)\big) = \frac{1}{6}(k+1)\big(2k^2 + 7k + 6\big) = \\ &= \frac{1}{6}(k+1)(k+2)(2k+3), \end{aligned}$$

as desired. The Principle of Mathematical Induction applies to get that p_n is true for all $n \geq 1$.

(b) $\frac{1}{2!}+\frac{2}{3!}+\frac{3}{4!}+\ldots+\frac{n-1}{n!}=\frac{n!-1}{n!}$, for all $n \geq 2$.

Solution: For an integer $n \geq 2$, define the statement

$$p_n: \quad \frac{1}{2!}+\frac{2}{3!}+\ldots+\frac{n-1}{n!}=\frac{n!-1}{n!}.$$

Observe that p_2 holds since

$$\frac{1}{2!}=\frac{2!-1}{2!}$$

Assume that p_k is true for some $k \geq 2$, and let us show that p_{k+1} is also true.

$$\frac{1}{2!}+\frac{2}{3!}+\ldots+\frac{k-1}{k!}+\frac{k}{(k+1)!} \stackrel{?}{=} \frac{(k+1)!-1}{(k+1)!}$$

Applying the induction hypothesis we obtain that

$$\frac{1}{2!}+\ldots+\frac{k-1}{k!}+\frac{k}{(k+1)!}=\frac{k!-1}{k!}+\frac{k}{(k+1)!}=\frac{\big(k!-1\big)(k+1)+k}{(k+1)!}=$$

$$=\frac{(k+1)!-(k+1)+k}{(k+1)!}=\frac{(k+1)!-1}{(k+1)!}$$

It proves that p_{k+1} is true and by the General Principle of Mathematical Induction we get that p_n is true for all $n \geq 2$.

(c) $\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\ldots\left(1-\frac{1}{n^2}\right)=\frac{n+1}{2n}$, for all $n \geq 2$. (It was in Test 1, 2021)

Solution: For $n \geq 2$, we define the statement

$$p_n: \quad \left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\ldots\left(1-\frac{1}{n^2}\right)=\frac{n+1}{2n}.$$

Let us first check that p_2 holds:

$$1-\frac{1}{2^2}=1-\frac{1}{4}=\frac{3}{4}=\frac{2+1}{2\cdot 2}.$$

Thus, p_2 holds. Assume now that p_k is true for some $k>2$, and let us show that p_{k+1} holds:

$$\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\ldots\left(1-\frac{1}{k^2}\right)\left(1-\frac{1}{(k+1)^2}\right)\stackrel{?}{=}\frac{(k+1)+1}{2(k+1)}$$

Applying the induction hypothesis (that is, p_k is true) we obtain that

$$\begin{aligned}&\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\ldots\left(1-\frac{1}{k^2}\right)\left(1-\frac{1}{(k+1)^2}\right)=\\&=\left(\frac{k+1}{2k}\right)\left(\frac{(k+1)^2-1}{(k+1)^2}\right)=\frac{k+1}{2k}\cdot\frac{k^2+2k}{(k+1)^2}=\frac{k(k+2)}{2k(k+1)}=\frac{k+2}{2(k+1)}\\&=\frac{(k+1)+1}{2(k+1)},\end{aligned}$$

which shows that p_{k+1} is true. To finish apply the Principle of Mathematical Induction applies to get that p_n is true for all $n \geq 2$.

(d) $(-1)^1 \cdot 1^2 + \ldots + (-1)^n \cdot n^2 = \frac{1}{2}(-1)^n \cdot n(n+1)$, for all $n \geq 1$. (It was in Test 1, 2022)

Solution: For $n \geq 1$, we define the statement

$$p_n : \quad (-1)^1 1^2 + (-1)^2 2^2 + \ldots + (-1)^n n^2 = \frac{1}{2}(-1)^n n(n+1).$$

Let us first check that p_1 holds:

$$(-1)^1 1^2 = -1$$

$$\frac{1}{2}(-1)^1 1(1+1) = -1.$$

Thus, p_1 holds. Next, we assume that p_k is true for some $k \geq 1$, and we have to show that p_{k+1} holds:

$$(-1)^1 1^2 + (-1)^2 2^2 + \ldots + (-1)^k k^2 + (-1)^{k+1}(k+1)^2 \stackrel{?}{=} \frac{1}{2}(-1)^{k+1}(k+1)(k+2).$$

Applying the induction hypothesis (that is, p_k is true) we obtain that

$$\begin{aligned}
&(-1)^1 1^2 + (-1)^2 2^2 + \ldots + (-1)^k k^2 + (-1)^{k+1}(k+1)^2 \\
&= \frac{1}{2}(-1)^k k(k+1) + (-1)^{k+1}(k+1)^2 = (-1)^k (k+1)\left(\frac{1}{2}k + (-1)(k+1)\right) = \\
&= (-1)^k (k+1)\frac{-k-2}{2} = \frac{1}{2}(-1)^k (k+1)(-1)(k+2) = \frac{1}{2}(-1)^{k+1}(k+1)(k+2),
\end{aligned}$$

which shows that p_{k+1} is true. By the Principle of Mathematical Induction we get that p_n is true for all $n \geq 1$.

2. Let a_n be the sequence given by $a_1 = 1$, $a_2 = 8$ and $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 3$. **Using strong induction prove** that

$$a_n = 3 \cdot 2^{n-1} + 2(-1)^n,$$

for all $n \in \mathbb{N}$.

Solution: For $n \geq 1$, we define the statement

$$p_n : \quad a_n = 3 \cdot 2^{n-1} + 2(-1)^n.$$

Notice that p_1 is true since

$$3 \cdot 2^{1-1} + 2(-1)^1 = 3 - 2 = 1 = a_1.$$

Take $k > 1$ and assume that $p_1, p_2, \ldots, p_k$ are true, and let us show that p_{k+1} is true. In particular, we have that p_k and p_{k-1} hold, and so

$$\begin{aligned} a_{k+1} = a_k + 2a_{k-1} &= 3 \cdot 2^{k-1} + 2(-1)^k + 2\big(3 \cdot 2^{k-2} + 2(-1)^{k-1}\big) \\ &= 3 \cdot 2^{k-1} + 2(-1)^k + 3 \cdot 2^{k-1} + 2 \cdot 2(-1)^{k-1} \\ &= 2 \cdot 3 \cdot 2^{k-1} + 2(-1)^k + 2 \cdot 2(-1)^{k-1} \\ &= 3 \cdot 2^k + 2(-1)^k\big(1 + 2(-1)^{-1}\big) \\ &= 3 \cdot 2^k + 2(-1)^{k+1}, \end{aligned}$$

which shows that p_{k+1} holds.

Notice that we needed to use Strong Induction here because in order to prove that p_{k+1} holds we applied both p_k and p_{k-1}.

3. **Apply the Euclidean Algorithm to compute** gcd(7854, 2145) **and express it as a linear combination of** 7854 **and** 2145**.**

Solution: We first apply the Division Algorithm until we find a zero remainder:

$$\begin{aligned} 7854 &= 3 \cdot 2145 + 1419 \\ 2145 &= 1 \cdot 1419 + 726 \\ 1419 &= 1 \cdot 726 + 693 \\ 726 &= 1 \cdot 693 + 33 \\ 693 &= 21 \cdot 33 + 0. \end{aligned}$$

Since 33 is the last remainder before the zero remainder, we can conclude that gcd(7854, 2145) = 33. Next, to write 33 as a linear combination of 7854 and

2145, we eliminate the remainders as follows:

$$\begin{array}{rclcrcl}
7854 &=& 3\cdot 2145 + 1419 & \Rightarrow & 1419 &=& 7854 + (-3)\cdot 2145 \qquad (1)\\
2145 &=& 1\cdot 1419 + 726 & \Rightarrow & 726 &=& 2145 + (-1)\cdot 1419 \qquad (2)\\
1419 &=& 1\cdot 726 + 693 & \Rightarrow & 693 &=& 1419 + (-1)\cdot 726 \qquad (3)\\
726 &=& 1\cdot 693 + 33 & \Rightarrow & 33 &=& 726 + (-1)\cdot 693 \qquad (4)
\end{array}$$

From here, we get that

$$\begin{aligned}
33 &\overset{(??)}{=} 726 + (-1)\cdot 693 \overset{(??)}{=} 726 + (-1)(1419 + (-1)\cdot 726) = (-1)1419 + 2\cdot 726\\
&\overset{(??)}{=} (-1)1419 + 2\cdot(2145 + (-1)\cdot 1419) = 2\cdot 2145 + (-3)1419\\
&\overset{(??)}{=} 2\cdot 2145 + (-3)\big(1\cdot 7854 + (-3)\cdot 2145\big) = (-3)\cdot 7854 + 11\cdot 2145
\end{aligned}$$

Thus: $\gcd(7854,\, 2145) = 33 = (-3)\cdot 7854 + 11\cdot 2145$.

4. **In each case determine whether the statement is true or false.** If true, provide an argument supporting your answer; if false, provide a counterexample. ((a)-(c) were in Test 1, 2021.)

   (a) Suppose that n and m are two integers not both zero, and $d > 1$ is an integer satisfying that $d = xm + yn$ for some integers x and y. Then $d = \gcd(n, m)$.

   (b) Suppose that n and m are two integers not both zero such that $1 = xm + yn$ for some integers x and y. Then $1 = \gcd(n, m)$.

   (c) If $d = \gcd(n, m)$ and $n = da$ and $m = db$. Then $\gcd(a, b) = 1$.

   (d) If $a = a_1c$ and $b = b_1c$, where $c = \gcd(a,\, b)$, then $\mathrm{lcm}(a,\, b) = a_1b_1c$.

Solution: (a) **False:** For example:

$$12 = 1\cdot 4 + 1\cdot 8$$

and 12 is not the gcd of 4 and 8, since $\gcd(4,\, 8) = 4$.

(b) **True:** It is enough to show that 1 is the only common positive divisor of m and n. To do so, suppose that $d \geq 1$ is such that $d|m$ and $d|n$. Then $d|(xm+yn) = 1$, which implies that $d = 1$.

(c) **True:** By Bézout's Lemma we can find integers x and y such that $d = xn+ym$. Now, we are assuming that $n = da$ and $m = db$, so

$$d = xn+ym = x(da)+y(db) = d(xa+yb) \Rightarrow d\Big(1-(xa+yb)\Big) = 0 \Rightarrow xa+yb = 1.$$

This implies that $\gcd(a, b) = 1$.

(d) **True:** We know that $ab = \gcd(a,\, b)\mathrm{lcm}(a,\, b)$. Keeping in mind that $a = a_1 c$, $b = b_1 c$ and $c = \gcd(a,\, b)$ we get that

$$(a_1 c)(b_1 c) = c\,\mathrm{lcm}(a,\, b),$$

which implies that $\mathrm{lcm}(a,\, b) = a_1 b_1 c$.

5. **Use the** ***Prime Factorisation Theorem*** **to find the** gcd **and the** lcm **of the following pair of numbers:**
   (a) $m = 139, \quad n = 278$
   (b) $m = 735, \quad n = 110$

Solution:

(a)
$$139 = 2^0 \cdot 139^1$$
$$278 = 2^1 \cdot 139^1$$

Thus $\gcd(139,\, 278) = 2^0 \cdot 139^1 = 139$ and $\mathrm{lcm}(139,\, 278) = 2^1 \cdot 139^1 = 278$.

(b)
$$735 = 5 \cdot 147 = 3^1 \cdot 5^1 \cdot 7^2 = 2^0 \cdot 3^1 \cdot 5^1 \cdot 7^2 \cdot 11^0$$
$$110 = 2^1 \cdot 5^1 \cdot 11^1 = 2^1 \cdot 3^0 \cdot 5^1 \cdot 7^0 \cdot 11^1$$

Thus $\gcd(735,\, 110) = 2^0 \cdot 3^0 \cdot 5^1 \cdot 7^0 \cdot 11^0 = 5$ and
$\mathrm{lcm}(735,\, 110) = 2^1 \cdot 3^1 \cdot 5^1 \cdot 7^2 \cdot 11^1 = 16170$.

6. Suppose that m and n are coprime and that $k|m$. Prove that k and n are coprime.

Solution: The plan is to apply Bézout's Lemma and the fact that it converse is true when the gcd is 1. Recall that the converse of Bézout's Lemma is not true in general.

From $\gcd(m, n) = 1$ we get that $sm + tn = 1$ for some $s, t \in \mathbb{Z}$. On the other hand from $k|m$ we get that $m = uk$ for some $u \in \mathbb{Z}$. Substituting m in $sm + tn = 1$ we get that

$$1 = sm + tn = s(uk) + tn = (su)k + tn,$$

which implies that $\gcd(k, n) = 1$.

7. **In each case determine whether the statement is true or false.** If true, provide an argument supporting your answer; if false, provide a counterexample. ((e) was in Test 1, 2021, while (f) was in Test 1, 2022.)

   (a) $40 \equiv 13 \pmod 9$

   (b) There are no integers a and b such that $a^2 - 3b = 2$.

   (c) $(51 + 68) \equiv 1 \pmod 5$.

   (d) $1111 \equiv 1 \pmod 2$.

   (e) Suppose that $p > 2$ is a prime. If $p \equiv 3 \pmod 4$, then $(p-1)/2$ is odd.

   (f) No integer of the form $n^2 + 1$ is a multiple of 7.

Solution:

(a) is **true** because $40 - 13 = 27 = 3 \cdot 9$, i.e., a multiple of 9.

(b) is **true:** Suppose that there exists integers a and b such that $a^2 - 3b = 2$. Then in $\mathbb{Z}_3$ we would have that $(\bar{a})^2 - \bar{3}\bar{b} = \bar{2}$, that is, $(\bar{a})^2 = \bar{2}$. But the equation $x^2 = \bar{2}$ has no solutions in $\mathbb{Z}_3$ since

$$\begin{aligned}
(\bar{0})^2 &= \overline{0^2} = \bar{0},\\
(\bar{1})^2 &= \overline{1^2} = \bar{1},\\
(\bar{2})^2 &= \overline{2^2} = \bar{4} = \bar{1}.
\end{aligned}$$

So, there are no integers satisfying that relation and the statement is true.

(c) is **false:** $(51 + 68) - 1 = 118$ and 5 does not divide 118.

(d) is **true:** $1111 - 1 = 1110$ and $2|1110$.

(e) is **true:** Suppose on the contrary that $(p-1)/2$ is even. Then $(p-1)/2 = 2k$ for some integer k. Thus $p - 1 = 4k$, which implies that $p - 1 \equiv 0 \pmod 4$, i. e., $p \equiv 1 \pmod 4$, a contradiction.

(f) is **True:** Consider the equation $x^2 + 1 \equiv 0 \pmod 7$, or equivalently, $x^2 \equiv 6 \pmod 7$. Notice that no element of $\mathbb{Z}_7$ satisfies such equation; in fact:

$$\begin{gathered}
\bar{0}^2 = \bar{0}, \quad \bar{1}^2 = \bar{1}, \quad \bar{2}^2 = \overline{2^2} = \bar{4}, \quad 3^2 = \overline{3^2} = \bar{9} = \bar{2}, \quad \bar{4}^2 = \overline{4^2} = \overline{16} = \bar{2},\\
\bar{5}^2 = \overline{5^2} = \overline{25} = \bar{4}, \quad \bar{6}^2 = \overline{6^2} = \overline{36} = 1.
\end{gathered}$$

Thus, there are no integers n such that $n^2 + 1$ is a multiple of 7.

8. Find the remainder of 6^{1001} when is divided by 7 (it was in Test 1, 2021).

Solution: Keeping in mind how the operations work in $\mathbb{Z}_7$, we first need to find d such that $(\bar{6})^d = 1$. Next, we will apply the Division Algorithm with 1001 and d to find integers q and r with $0 \leq r < d$ such that

$$1001 = q \cdot d + r,$$

and so $(\bar{6})^{1001} = (\bar{6})^{q \cdot d + r} = \left((\bar{6})^d\right)^q (\bar{6})^r = (\bar{1})^q (\bar{6})^r = (\bar{6})^r$, and we would have already computed $(\bar{6})^r$ since $r < d$. So, let us compute the powers of $\bar{6}$ in $\mathbb{Z}_7$:

$$(\bar{6})^1 = \bar{6},$$
$$(\bar{6})^2 = \overline{6^2} = \overline{36} = \bar{1}.$$

Now, we apply the Division Algorithm with 1001 and 2 to obtain

$$1001 = 500 \cdot 2 + 1.$$

In $\mathbb{Z}_7$ we then have:

$$(\bar{6})^{1001} = (\bar{6})^{500 \cdot 2 + 1} = \left((\bar{6})^2\right)^{500} \cdot (\bar{6})^1$$

Using that $(\bar{6})^2 = \bar{1}$, we obtain that $(\bar{6})^{1001} = \left((\bar{6})^2\right)^{500} \cdot (\bar{6})^1 = \bar{6}$.

9. (a) **In $\mathbb{Z}_{20}$, find the inverse of $\overline{11}$ and use it to solve $\overline{11}x = \overline{16}$.**
   (b) **In $\mathbb{Z}_{15}$, find the inverse of $\bar{4}$ and use it to solve the equation $\bar{4}x = \overline{12}$.** (It was in Test 1, 2021).

Solution: (a) From $\gcd(11, 20) = 1$ we can conclude that $\overline{11}$ has an inverse in $\mathbb{Z}_{20}$. Such inverse is the coefficient of 11 in Bézout's Identity. To find it, we apply the Euclidean Algorithm:

$$20 = 1 \cdot 11 + 9,$$
$$11 = 1 \cdot 9 + 2,$$
$$9 = 4 \cdot 2 + 1.$$

We have that

$$1 = 9 - 4 \cdot 2 = 9 - 4 \cdot (11 - 1 \cdot 9) = (-4)11 + 5 \cdot 9 = (-4)11 + 5(20 - 1 \cdot 11) =$$
$$= 5 \cdot 20 + (-9)11,$$

which implies that $\overline{11}^{-1} = \overline{-9} = \overline{11}$, that is, $\overline{11}$ is self-inverse. We solve now the equation $\overline{11}x = \overline{16}$ in $\mathbb{Z}_{20}$. To do so, multiply both sides by $\overline{11}^{-1} = \overline{11}$:

$$\left(\overline{11}^{-1} \cdot \overline{11}\right)x = \overline{11}^{-1} \cdot \overline{16} \;\Rightarrow\; x = \overline{11 \cdot 16} = \overline{176} = \overline{16},$$

since $176 - 16 = 160$, which is a multiple of 20.

(b) Notice that $\bar{4}$ has an inverse in $\mathbb{Z}_{15}$ since $\gcd(4, 15) = 1$.

An application of Bézout's Lemma yields $4u + 15v = 1$, for some integers u, v. In $\mathbb{Z}_{15}$ we have that $\overline{15} = \bar{0}$, so

$$\bar{1} = \overline{4u + 15v} = \bar{4} \cdot \bar{4} + \overline{15} \cdot \bar{v} = \bar{4} \cdot \bar{u}.$$

This says that the inverse of $\bar{4}$ in $\mathbb{Z}_{15}$ is the class of the coefficient of 4 obtained from Bézout's Lemma. To find u we need to apply the Euclidean Algorithm:

$$15 = 3 \cdot 4 + 3 \tag{5}$$
$$4 = 1 \cdot 3 + 1 \tag{6}$$
$$3 = 3 \cdot 1 + 0$$

We now express 1 as a linear combination of 15 and 4:

$$1 \overset{(2)}{=} 4 + (-1) \cdot 3 \overset{(1)}{=} 4 + (-1) \cdot \big(15 + (-3) \cdot 4)\big) = (-1) \cdot 15 + 4 \cdot 4$$

So, the inverse of $\bar{4}$ in $\mathbb{Z}_{15}$ is $\bar{4}$.

To solve the equation $\bar{4}x = \overline{12}$, we multiply both sides by $\bar{4}$ (the inverse of $\bar{4}$ in $\mathbb{Z}_{15}$:

$$x = \bar{4} \cdot \overline{12} = \overline{4 \cdot 12} = \overline{48} = \bar{3},$$

since $48 - 3 = 45$ and $5|45$.

10. Consider the decimal representation $n = d_j d_{j-1} \dots d_2 d_1 d_0$ of a positive integer n. Prove that $3|n$ if and only if $3|(d_0 + d_1 + \dots + d_j)$. (It was in Test 1, 2022.)

Just in case you need an example of the decimal representation: for $n = 3785$ we have that $d_0 = 5$, $d_1 = 8$, $d_2 = 7$ and $d_3 = 3$.

Solution: Let us begin by noticing that $3|n$ if and only if $n \equiv 0 \pmod 3$. So, we need to show the following:

$$n \equiv 0 \pmod 3 \Leftrightarrow 3|(d_0 + d_1 + \dots + d_j)$$

To do so, notice that $n = d_j d_{j-1} \dots d_2 d_1 d_0 = d_0 + 10d_1 + 10^2 d_2 + \dots + 10^j d_j$. On the other hand from $10 \equiv 1 \pmod 3$, we obtain that

$$n = d_0 + 10d_1 + 10^2 d_2 + \dots + 10^j d_j \equiv (d_0 + d_1 + \dots + d_j) \pmod 3. \quad (7)$$

Using (**??**) the result follows trivially. In fact:

If $n \equiv 0 \pmod 3$, then $d_0 + d_1 + \dots + d_j \equiv 0 \pmod 3$ by (**??**), which means that $3|(d_0 + d_1 + \dots + d_j)$. Conversely, if $3|(d_0 + d_1 + \dots + d_j)$, then $d_0 + d_1 + \dots + d_j$ is a multiple of 3, and so it has zero remainder when divided by 3, that is, $d_0 + d_1 + \dots + d_j \equiv 0 \pmod 3$ and $n \equiv 0 \pmod 3$ by (**??**).

11. Find all the solutions to the simultaneous congruences: ((b) was in Test 1, 2022)

   (a) $x \equiv 1 \pmod 5$ and $2x \equiv 1 \pmod 9$

   (b) $3x \equiv 2 \pmod 5$ and $x \equiv 2 \pmod 3$

Solution: (a) We proceed like in the proof of the Chinese Remainder Theorem, and begin by looking the solutions of the congruence $x \equiv 1 \pmod 5$; which are the following:

$$x = 1 + 5k, \quad k \in \mathbb{Z}.$$

Take $k \in \mathbb{Z}$ and make $x = 1 + 5k$ in the second congruence to get that

$$2(1+5k) \equiv 1 \pmod 9 \;\Leftrightarrow\; 2 + 10k \equiv 1 \pmod 9 \;\Leftrightarrow\; 10k \equiv -1 \equiv 8 \pmod 9$$

But $\overline{10} = \bar{1}$ in $\mathbb{Z}_9$, so the equation above becomes $x \equiv 8 \pmod 9$. From here, we obtain that a solution of the given congruences is $x = 1 + 5 \cdot 8 = 41$. By the Chinese Remainder Theorem we can conclude that all the solutions are the following:

$$41 + (5 \cdot 9)n = 41 + 45n, \; n \in \mathbb{Z}.$$

(b) Proceeding like in the proof of the Classical Chinese Remainder Theorem, we first look at the solutions of $x \equiv 2 \pmod 3$, which are of the form $x = 2 + 3k$, for $k \in \mathbb{Z}$. Sustituing in the first congruence we obtain:

$$3(2+3k) \equiv 2 \pmod 5 \;\Leftrightarrow\; 9k \equiv -4 \pmod 5$$

Next, since $9 \equiv 4 \pmod 5$ and $-4 \equiv 1 \pmod 5$ our equation becomes:

$$4k \equiv 1 \pmod 5$$

To do so, just notice that $16 \equiv 1 \pmod 5$, that is, $\bar{4}^{-1} = \bar{4}$ in $\mathbb{Z}_5$. Using this we get that $k \equiv 4 \pmod 5$. Thus, a common solution of the simultaneous congruences is $x = 2 + 3 \cdot 4 = 14$, and by the Chinese Theorem Remainder we can conclude that all the solutions are:

$$14 + (3 \cdot 5)n = 14 + 15n, \; n \in \mathbb{Z}.$$